Bike Sharing Demand Prediction

Random forest and regression model comparison

Problem statement
Estimate the number of bicycle rentals in an hour from calendar, weather, and normalized weather measurements. A reliable estimate can help a bike-share operator anticipate hourly fleet demand.

Dataset
The UCI Bike Sharing Dataset (hour.csv) contains 17,379 hourly records. It provides 12 predictors used here and the numeric target cnt (rental count). The target median is 142 rentals per hour. The original casual and registered counts are deliberately excluded because they are components of cnt and would leak the answer.

Preprocessing
Missing values among selected columns: 0; duplicate rows: 0. Numeric weather inputs are median-imputed and standardized. Calendar and weather-category codes are imputed with the most frequent value and one-hot encoded. Preprocessing is inside each model Pipeline, so each validation fold learns transformations from its training fold.

Training and model selection
Data was divided 80/20 with random_state=42. The models are Random Forest (nonlinear tree ensemble) and Ridge Regression (regularized linear baseline). Five-fold shuffled cross-validation on the training set selected the model with the lowest mean RMSE: Random forest (61.76 rentals). The test set was reserved for final evaluation.

Dataset source: UCI Machine Learning Repository, Bike Sharing Dataset (Fanaee-T and Gama, 2013).

Evaluation results

Model	CV RMSE	Test MAE	Test RMSE	Test R^2
Random forest	61.76	38.89	58.35	0.892
Ridge regression	102.42	74.06	100.37	0.682

Selected model: Random forest
Test MAE: 38.89 rentals; test RMSE: 58.35; test R^2: 0.892.

MAE is typical absolute error; RMSE penalizes large misses and is the cross-validation selection measure. R^2 measures test-set fit. Classification scores do not apply to regression.

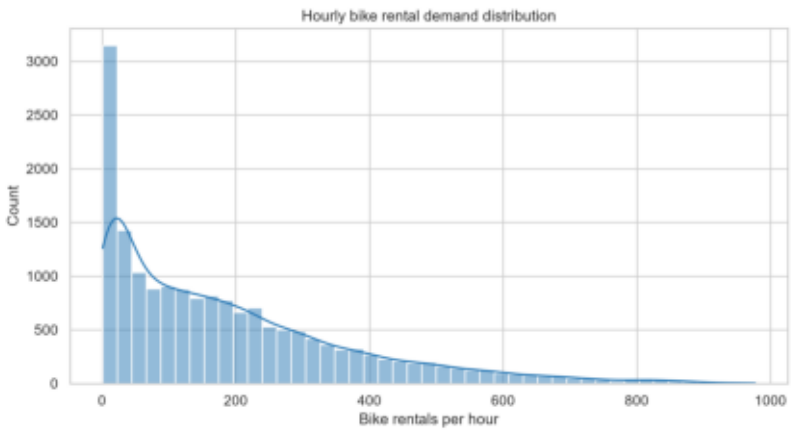


Figure 1. Distribution of hourly rental counts.

Patterns and predictions

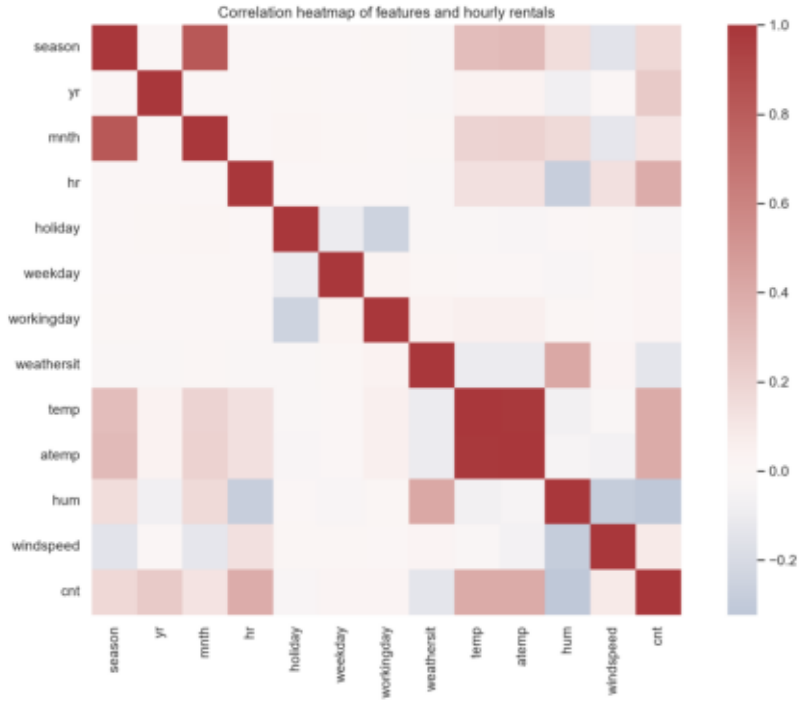


Figure 2. Correlations show pairwise linear associations; they do not imply causation.

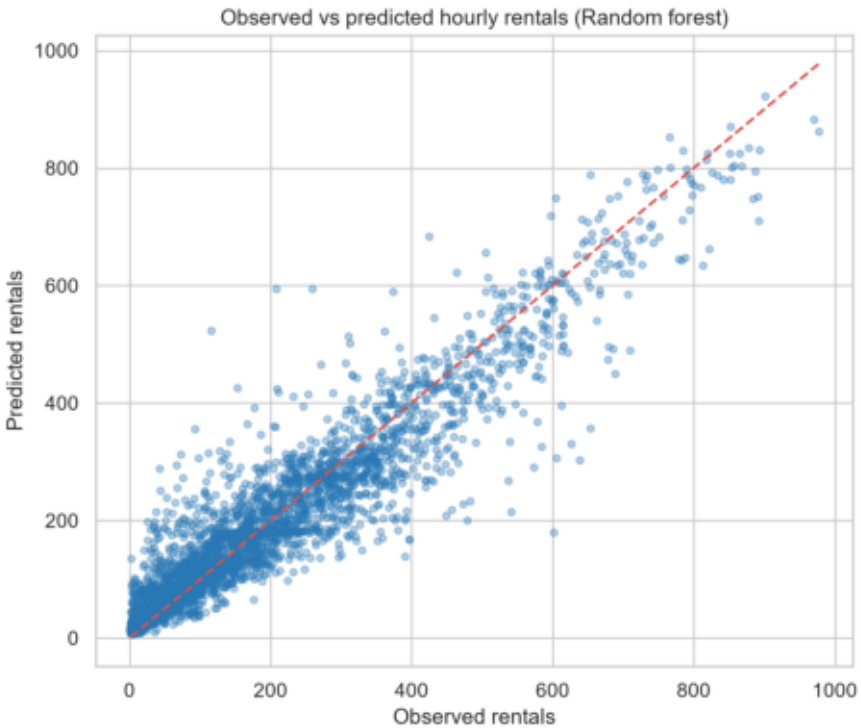


Figure 3. Points near the diagonal indicate closer predictions; spread indicates error.

Analysis, limitations, and course outcome

Model comparison
Random forest had the lowest mean five-fold validation RMSE (61.76) and was selected without using test-set scores. Model-comparison graph error bars show variation in RMSE across the five folds. The selected model's test error is 58.35 rentals RMSE and 38.89 rentals MAE.

Understanding errors
An underprediction can leave too few bicycles available; an overprediction can leave bikes underused. The costs depend on station, time, and rebalancing capacity. For operations, MAE is an intuitive typical error, while RMSE is useful when large demand misses are especially costly. A real deployment should also evaluate peak-hour or station-specific errors and choose metrics based on operational costs.

Limitations
This is historical data from a limited period and region; demand patterns can change. The dataset has no station-level availability, event, service disruption, or real-time information. Random splitting can place nearby hours in both train and test sets, so it may overstate performance on future unseen time periods. A chronological holdout and external validation are recommended before operational use. Predictions are estimates, not guarantees.

CO5 – Solve real life problems using appropriate machine learning models and evaluate the performance measures
This project translates hourly bike allocation into a supervised regression problem, compares suitable models using five-fold cross-validation, evaluates the selected approach on a held-out test set, visualizes the results, and discusses how prediction errors affect the real-world decision.

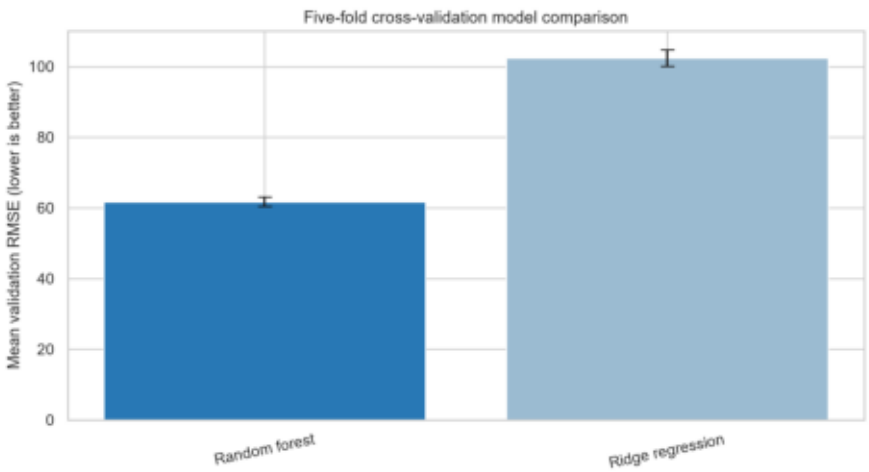


Figure 4. Five-fold cross-validation RMSE; lower is better.